

Appendix A: Study Criteria, Methodology and Assumptions

A.1 Study Criteria

ATC used NERC Reliability Standard TPL-001-4, MISO BPM-015-r22, ATC Planning Criteria (PLG-CR-0001-V20.2), ATC Transmission Planning Assessment Practices (PLG-METH-0002-V20.1) and ATC Generating Facility Interconnection Guide (Revision 11.0) to evaluate the reliability impact of each generator project on the transmission system. They can be found in Appendix B.

A.1.1 Steady State Injection Constraints

For NERC Category P0-P7 contingency evaluation, ATC adheres to the MISO defined generation injection constraints (BPM-015) including:

- i. The generator has a larger than twenty percent (20%) sensitivity factor on the overloaded facilities under post contingent condition (see NERC TPL) or five percent (5%) sensitivity factor under system intact condition, or
- ii. The overloaded facility or the overload-causing contingency is at generator's outlet, or
- iii. The megawatt impact due to the generator is greater than or equal to twenty percent (20%) of the applicable rating (normal or emergency) of the overloaded facility, or
- iv. For any other constrained facility, where none of the Study Generators meet one of the above criteria in i, ii or iii, however, the cumulative megawatt impact of the group of study generators is greater than twenty percent (20%) of the rating of the facility, then only those study generators whose individual megawatt impact is greater than five percent (5%) of the rating of the facility and has Distribution Factor greater than five percent (5%) (PTDF or OTDF) will be responsible for mitigating the cumulative megawatt impact constraint, or
- v. Impacts on Affected Systems would be classified as Injection constraints based on the Affected Systems' criteria
- vi. Any other applicable Transmission Owner FERC filed Local Planning Criteria.

A.1.2 Steady State Voltage Constraints

For NERC Category P0-P7 contingency evaluation, ATC adheres to the MISO defined generation injection constraints (BPM-015) including:

Bus voltages outside of the defined limits (based on ATC Planning Criteria) in the post case will be recorded and compared to the Base Case values. Bus voltages will be considered voltage constraints if, for a given contingency, the bus voltage is outside of the allowed voltage range for the post case and is at least 0.01 per unit worse than the Base Case voltage for the same contingency.

Power flow solutions must converge for contingencies in the electrical proximity of the studied Generator. Divergence of a power flow solution indicates potential voltage collapse. A "fix" must be identified for any non-converging power flow simulation and may include generator operating restrictions. [Note: Non-convergence may be due to solution settings such as switched shunt operation and/or LTC action.]

A.1.3 Stability Criteria

ATC stability criteria applicable to this study from ATC Planning Criteria PLG-CR-00-V20.2 include:

1. For generator transient stability, faults will be modeled on the high side bus at generating plants.
2. For generating units with actual “as built” or “field setting” dynamic data, a 0.5 cycle margin will be added to the expected clearing time (ECT) for dynamic Contingency simulations. For generating units with assumed, typical, or proposed dynamic data, a 1.0 cycle margin will be added to the ECT for dynamic Contingency simulations. The ECT margin will only be added to near end protective device for normal clearing faults (i.e. P1, P2, P3, P6 and P7) or it will be only added to the protective device with delayed clearing (i.e. P4 and P5). The total clearing time (ECT + margin) must be equal to or less than the calculated critical clearing time (CCT) from the simulation.
3. Generator transient stability will be demonstrated for at least one key Contingency for each applicable P1 through P7 Contingency. Unacceptable transient stability performance occurs when any of the stability assessment criteria are not met.

A. Angular Stability Assessment

- i. Generating unit loses synchronism with the transmission system, unless it is deliberately islanded
- ii. Cascading tripping of transmission lines, tripping of transmission transformers or uncontrolled loss of load
- iii. Poorly damped angular oscillations where acceptable damping is defined in Section 1.2.2 of ATC Planning Criteria.

B. Voltage Stability Assessment

- i. Transient stability voltage response at applicable Bulk-Electric System (BES) buses serving load shall recover to at least 80% of nominal voltage within 2 seconds after the initiating event is cleared for all P1-P7 Contingencies
- ii. For voltage swings subsequent to fault clearing and the first voltage recovery above 80%, voltage dips at each applicable BES bus serving load for all P1-P7 Contingencies:
 - a. Shall not dip below 70% of nominal voltage for more than 30 cycles continuous, or
 - b. Shall not dip below 80% of nominal voltage for more than 2 seconds continuous.

C. Inverter-Based Resources Stability Assessment

Inverter-based resources refer to generating and energy storage resources that are asynchronously connected to the grid and are either completely or partially interfaced with the transmission system through power electronics.

- i. An inverter-based resource shall not trip for any studied Planning Events, unless it is deliberately islanded. This also applies to single line to ground (SLG) faults for Planning Events that are defined as three phase faults (e.g. certain P1, P3 and P6 events).
- ii. An inverter-based resource’s active power and reactive power as measured at the machine bus and bus voltage as measured at the Point-of-Interconnection (POI) shall be positively damped and meet one of the following two criteria:
 - a. Time domain analysis indicates a 50% or greater reduction in oscillation magnitude over the last four oscillation periods.
 - b. Peak-to-peak magnitudes during the last two seconds of the 20 second simulation shall not exceed 3% of their rated values (machine MVA base for active and reactive power and base kV at POI for voltage).

- iii. Inverter-based resources shall not re-enter fault ride-through mode more than once in the time period beginning 6 cycles after the fault clears until the end of the 20 second simulation.
- iv. Inverter-based resources should be designed and configured to continue current injection (active, reactive, or a combination of current) inside the “No Trip Zone” of the frequency and voltage ride-through curves of the currently effective version of PRC-024 unless a reliability study identifies a system need to cease injecting current. Inverter-based resources should be designed and configured to use momentary cessation only outside the “No Trip Zone” if this helps mitigate potential tripping conditions based on interconnection studies. Any use of momentary cessation should be based on equipment limitations or based on reliability studies identifying a system need. Return from momentary cessation upon voltage recovery to within the continuous operation range should occur as quickly as possible with no intentional time delay while maintaining stability.

Exceptions to these criteria (item ii, iii) given within this section may be considered on a case-by-case basis (e.g. for existing inverter-based resources or if NERC provides more specific inverter-based resource performance guidelines before ATC revises these criteria).

- 4. Where needed system reinforcement cannot be implemented in an appropriate timeframe, then a corrective plan must be established in order to respect System Operating Limits and/or Interconnected Reliability Operating Limits. Where appropriate, corrective plans may include generator redispatch, operating guides, and/or Special Protection Systems.

A.1.4 Transformer Initial Energization Criteria

Initial transformer energization must not result in voltages below 0.70 pu at the Point of Interconnection (POI). The energization was evaluated using the equation $V_{sag} = X / (X + 2.5 \cdot X_t)$ where X is the source reactance and X_t is the impedance of the transformer being energized. This equation is taken from the paper "Assessing and Limiting Impact of Transformer Inrush Current on power Quality" by Nagpal, M. T., T. G. Martinich, A. Moshref, K. Morison and P. Kundar, as published in the IEEE Transactions on Power Delivery, April 2006 (Vol. 21, No. 2, pp 890-896).

The source reactance (X in the equation above) is calculated from the three-phase fault current at the POI with the system configured as it is expected to be prior to the transformer energization. The reactance in ohms is calculated using the equation $X = V_{LL} / (A \cdot \sqrt{3})$ where V_{LL} is the line to line voltage and A is the fault current. A 115% multiplier is added to the conservative raw minimum voltage calculated using this methodology to obtain a more realistic minimum voltage.

A.2 Study Methodology and Assumptions

The results of this study are subject to change. The results of the study are based on data provided by the Interconnection Customers and other ATC system information that was available at the time the study was performed, and the injection study does not guarantee deliverability to the MISO energy market. If there are any significant changes in the generator and controls data, earlier queued Generator Interconnection Requests, related Transmission Service Requests, or ATC transmission system development plans, then the results of this study may also change significantly. Therefore, these requests are subject to restudy. The Interconnection Customers are responsible for communicating any significant Generating Facility data changes in a timely fashion to MISO and ATC at any point in time such changes are identified.

A.2.1 Steady State AC Power Flow Analysis

The steady state thermal and voltage AC power flow analysis were performed using a combination of PowerGem's TARA version 2001.1 and the Power Flow module of the Power System Simulation/Engineering program (PSS/E version 34) from PTI. These programs are accepted industry-wide for power flow analysis. All distribution factors were calculated using PowerGem's TARA version 2001.1.

This analysis was used to evaluate the intact system (NERC TPL-001-4 Category P0), single contingencies (NERC TPL-001-4 Category P1-P2) and certain multiple contingency conditions (NERC TPL-001-4 Category P4-P7). All AC power flow solutions were screened at 95% of actual equipment ratings in MVA.

The key solution options chosen for performing the TARA power flow solution activity are listed below.

	Base Case	AC Contingency Analysis
Area Interchange	Disabled	
Solution Method	Fast Decoupled	
Bus Mismatch Tolerance (MVA)	1	3
Tap Adjustments	Enabled	
PAR Adjustments	Enabled	
Shunt Adjustments	All Enabled	
DC Tap Adjustments	Enabled	
Generator VAR Limit Checking	0 Iterations	
Ratings	RATEA	RATEB
Loading Cutoff	95%	
Flow Change Cutoffs	5 MW/2% Rating	
Voltage Cutoffs	0.005 PU	

Note, all models were built using Power System Simulation/Engineering (PSS®E, Version 34) program from Siemens Power Technologies, Inc. (PTI).

A.2.1.1 Contingencies Studied

Table A.2.1-1 below provided a detailed listing of the NERC Planning Events simulated for the steady state analysis according to the NERC TPL-001-4 standard. The "contingency study area" referred in Table A.2.2.1 includes all 69 kV and above transmission facilities within ATC and selected CE, XCEL, MP, and DPC area contingencies. NERC Category P3 and P6 contingencies are only evaluated in DPP Phase 2 and Phase 3 system impact studies. NERC Category E contingencies (NERC TPL-001-4 Extreme Events) were not evaluated in the DPP system impact studies.

Table A.2.1-1 Contingencies Studied in Steady State Analysis

Category	Initial Condition	Event	Description
P0	Normal System	All Facilities in Service	P0 conditions were studied.
P1 Single Contingency	Normal System	Loss of one of the following: 1. Generator 2. Transmission Circuit 3. Transformer 4. Shunt Device 5. Single Pole DC Line	All applicable P1 contingencies in “contingency study area” were studied.
P2 Single Contingency	Normal System	1. Opening of a line section w/o a fault 2. Bus Section Fault 3. Internal Breaker Fault (non-Bus-tie Breaker) 4. Internal Breaker Fault (Bus-tie Breaker)	All applicable P2 contingencies in “contingency study area” were studied.
P3 Multiple Contingency	Loss of generator unit followed by System adjustments	Loss of one of the following: 1. Generator 2. Transmission Circuit 3. Transformer 4. Shunt Device 5. Single Pole DC Line	P3 contingencies in ATC system plus Stone Lake 345/161 kV transformer.
P4 Multiple Contingency (Fault plus stuck breaker)	Normal System	Loss of multiple elements caused by a stuck breaker (non-Bus-tie Breaker) attempting to clear a Fault on one of the following: 1. Generator 2. Transmission Circuit 3. Transformer 4. Shunt Device 5. Bus Section 6. Loss of multiple elements caused by a stuck breaker (Bus-tie Breaker) attempting to clear a Fault on the associated bus	All applicable P4 contingencies in “contingency study area” were studied.
P5 Multiple Contingency Fault plus relay failure to operate)	Normal System	Delayed Fault Clearing due to the failure of a non-redundant relay protecting the Faulted element to operate as designed, for one of the following: 1. Generator 2. Transmission Circuit 3. Transformer 4. Shunt Device 5. Bus Section	All applicable P5 contingencies in “contingency study area” were studied.

Category	Initial Condition	Event	Description
P6 Multiple Contingency (Two overlapping singles)	Loss of one of the following followed by System adjustments.	Loss of one of the following:	P6 contingencies in ATC system plus Stone Lake 345/161 kV transformer.
	1. Transmission Circuit 2. Transformer 3. Shunt Device 4. Single pole of a DC line	1. Transmission Circuit 2. Transformer 3. Shunt Device 4. Single pole of a DC line	
P7 Multiple Contingency (Common Structure)	Normal System	The loss of: 1. Any two adjacent (vertically or horizontally) circuits on common structure 2. Loss of a bipolar DC line	All applicable P7 contingencies in “contingency study area” were studied.

A.2.1.2 Monitored System

All 69 kV and above transmission facilities within ATC, XCEL, MP, and DPC were monitored for thermal constraints. All 69 kV and above transmission facilities within ATC were monitored for voltage constraints.

A.2.1.3 Generator Steady State Operating Restriction Study Methodology

The purpose of the study is to identify potential operating restrictions for study generators under prior outage conditions and raise awareness of these potential operational restrictions to customers.

Steady-state thermal constraints due to NERC Category P6 events (N-1-1) will be mitigated in the day-ahead and real-time market through the MISO binding constraint and other operating procedures. For MISO DPP system impact study purposes, a conservative approach as demonstrated in the following examples is used to identify potential steady-state operating restrictions for study generators.

Example 1 – A Single Study Generator Responsible for a Category P6 Injection Constraint

A999 is a 100 MW wind generation request, dispatched at 15.6 MW in a summer peak ERIS study case. Line XY is rated 120 MVA Emergency and is loaded to 125 MVA under a Category P6 event in the summer peak study case. A999 has a 25% distribution factor on line XY.

Potential A999 MW Reduction Required = the **smaller** of the two numbers below

1. $[\text{Study Case Contingency Flow} - \text{Rating}] / \text{Distribution Factor} = [125 - 120] / 0.25 = 20 \text{ MW}$
2. Pgen in the Summer Peak Study Case = 15.6 MW

Max MW Allowed :

Based on this study model, for this Category P6 event, the maximum MW allowed (or worst-case potential operational restriction) for A999 is 0 MW (i.e. complete shutting down the unit) after the 1st outage.

Example 2 – Multiple Study Generators Responsible for a Category P6 Injection Constraint

A999 is a 100 MW wind generation request, dispatched at 15.6 MW in a summer peak ERS study case. B999 is a 50 MW solar generation request, dispatched at 50 MW in the summer peak study case. Line XY is rated 120 MVA Emergency and is loaded to 125 MVA under a Category P6 event in the Shoulder Study Case. A999 has a 25% distribution factor on line XY. B999 has a 20% distribution factor on line XY.

Potential A999 MW Reduction Required = the **smaller** of the two numbers below

1. [Study Case Contingency Flow – Rating] / Distribution Factor = $[125 - 120] / 0.25 = 20$ MW
2. Pgen in the Shoulder Study Case = 15.6 MW

Potential B999 MW Reduction Required = the **smaller** of the two numbers below

3. [Study Case Contingency Flow – Rating] / Distribution Factor = $[125 - 120] / 0.20 = 25$ MW
4. Pgen in the Shoulder Study Case = 50 MW

Max MW Allowed :

Based on this study model, for this Category P6 event, the maximum MW allowed (or worst-case potential operational restriction) for A999 is 0 MW (i.e. complete shutting down the unit) after the 1st outage.

Based on this study model, for this Category P6 event, the maximum MW allowed (or worst-case potential operational restriction) for B999 is 25 MW after the 1st outage.

Notes:

- The steady-state Category P6 operating restriction study should be performed after final ERS Network Upgrades and NRIS Network Upgrades have been identified. The study should be performed on study models with final ERS Network Upgrades and NRIS Network Upgrades included.
- The Max MW Allowed is only calculated for Category P6 Constraints that meet MISO's Injection Constraints criteria and for study generators that have positive distribution factors on those constraints.
- The Max MW Allowed is calculated assuming a single unit is redispatched (i.e. no other units have output reduced). This approach provides ATC Operations and Generator Owners the worst potential operational restrictions based on study models.
- If multiple P6 (N-1-1) events result in the same Max MW Allowed reduction requirement, only the worst N-1-1 event (highest loading % on the constraint) will be summarized in the SIS report.

A.2.2 Dynamic Stability Analysis

The stability and grid disturbance performance analysis were performed using Transient Security Assessment Tool (TSAT) version 19.0. This program is accepted industry-wide for dynamic stability analysis.

The system was tested for stability constraints after the addition of all generators in the current queue cycle. If any constraints were found, the fault would also be simulated in the case representing the system before the introduction of the current queue cycle. Any constraints occurring with the requested

interconnections but not without, as well as constraints that get worse after their addition, are attributed to the current queue cycle in this report.

A.2.2.1 Contingencies Studied

For stability analysis, a set of facilities in the vicinity of the generator/power plant of concern are selected as contingencies, based on engineering judgment and ATC TPL-001-4 Contingencies Rationale (PLG-RTNL-0005-V1). Generator transient stability will be demonstrated for at least one key contingency for each applicable NERC Category P1, P4, P5, P6, and P7. (Applicable NERC Standards: TPL-001-4-R2.4, R2.5, R4; ATC Planning Criteria Section 1.2).

A.2.2.2 Monitored System

The area local to the generators under study will be monitored, based on engineering judgement, using TSAT. TSAT monitoring may include:

- Angles of the generators under study and select local generators.
- Real power output of the generators under study and select local generators.
- Reactive power output of the generators under study and select local generators.
- Voltage responses of plant buses under study and select local plants.
- Voltage responses of select transmission buses near the generator under study.

A.2.3 Short Circuit Analysis

Short Circuit analysis will be performed using the ATC Protection models with the CAPE (Computer-Aided Protection Engineering) simulation software. ATC CAPE models include all ATC-owned substations, all ATC local distribution customer owned substations, as well as limited buses into ATC neighboring systems (MP, ComEd, XCEL, DPC, ITC). Two study models will be developed:

1. Before Case – This will be the 5-Year CAPE category (Near-Term Planning Horizon) with all existing and prior queue generators modeled and dispatched. All prior queue ERIS and NRIS Network Upgrade projects that change system impedance will also be modeled.
2. After Case - This will be the Before Case plus all current queue study generators dispatched and final current queue ERIS and NRIS Network Upgrade projects that change system impedance modeled.

Maximum fault duty, minimum fault duty, maximum fault Thevenin equivalent, and minimum fault Thevenin equivalent will all be calculated at the POI of each request using After Case.

Single line-to-ground (SLG) and Three phase-to-ground (3PG) faults will be simulated in Before and After Cases and the results will be compared.

Breaker interrupting duty for ATC protective devices will be compared to breaker interrupting ratings and any breaker exceeding 100% of the interrupting rating due to the addition of study generators in After Case is required to be upgraded.

A.2.4 Weak Grid Analysis

The strength of the grid is determined by its impedance and mechanical inertia i.e. kinetic energy stored in the rotating parts of the connected generator. Short Circuit Ratio (SCR) is not a strength indicator of the entire system but a measure of the system strength at the specified bus or POI in this study. The SCR of a bus is an indication of the strength of the bus i.e. whether the bus can maintain its voltage in response to reactive power variations. A system having high SCR will experience less change in bus voltage than a network with low SCR.

NERC [1] defines SCR as the ratio between short circuit apparent power (SCMVA) from a 3LG fault at a given location in the power system to the rating of the inverter based resource connected to that location.

$$SCR_{POI} = \frac{SCMVA_{POI}}{MW_{VER}}$$

Another important measure of grid strength is Weighted SCR (WSCR). This method was devised by ERCOT [2] to calculate SCR for their Panhandle region which lacks conventional generation or significant load. This method considers the inter plant oscillations and tends to be a better indicator of the SCMVA.

The WSCR can be calculated as follows:

$$WSCR = \frac{\text{Weighted } S_{SCMVA}}{\sum_i^N P_{RMWi}}$$

$$WSCR = \frac{(\sum_i^N S_{SCMVAi} * P_{RMWi}) / \sum_i^N P_{RMWi}}{\sum_i^N P_{RMWi}}$$

$$WSCR = \frac{\sum_i^N S_{SCMVAi} * P_{RMWi}}{(\sum_i^N P_{RMWi})^2}$$

Where S_{SCMVAi} is the short circuit capacity at bus i without current contribution from nonsynchronous generation, P_{RMWi} is the MW output of the nonsynchronous generation to be connected at bus i . N is the number of wind plants fully interacting with each other and i is the nonsynchronous plant index.

ATC evaluates for weak grid conditions in the shoulder load case. The SCR will be evaluated with system in-tact and n-1 conditions. ATC will also evaluate WSCR as appropriate for system in-tact and n-1 conditions. The grouping of generators for WSCR will be based on engineering judgement. If ATC determines by engineering judgement that the addition of new inverter-based generation in an area results in a sufficiently weak system a PSCAD study will be required.

References:

[1] June 2017 NERC Reliability Guideline titled "Integrating Inverter Based Resources into Weak Power Systems: http://www.nerc.com/pa/RAPA/rg/ReliabilityGuidelines/Reliability_Guideline_-_Integrating_VER_into_Weak_Power_Systems.pdf

[2] Y. Zhang, S. H. F. Huang, J. Schmall, J. Conto, J. Billo and E. Rehman, "Evaluating system strength for large-scale wind plant integration," 2014 IEEE PES General Meeting, National Harbor, MD, 2014, pp. 1-5.

A.2.5 PRC-024-2 Analysis

Introduction

PRC-024-2 is a NERC standard that applies to Generator Owners. The standard requires that Generator Owners set their frequency relays such that generating units do not trip within the “no trip zone” defined in Attachment 1 of the standard. PRC-024-2 also requires that Generator Owners set their voltage relays such that generating units do not trip within the “no trip zone” defined in Attachment 2 of the standard. Although this standard applies to Generator Owners, ATC will evaluate adherence to PRC-024-2 for each interconnection request studied in the MISO DPP process and provide results in the DPP System Impact Study.

The NERC Reliability Guideline on Inverter-Based Resource Performance, states momentary cessation should not be used within the voltage and frequency ride through curves specified in PRC-024-2. ATC will evaluate the results from this analysis for momentary cessation

https://www.nerc.com/comm/%E2%80%8C%20Reliability_Guidelines_DL/Inverter-Based_Resource_Performance_Guideline.pdf

Evaluating Frequency Protective Relay Settings

Frequency relays will not be evaluated in a PSSE dynamics simulation because of known limitations of frequency representation in the PSSE software. The frequency is calculated as the derivative of the voltage angle at any given bus. When a fault is applied to a bus the voltage angle changes rapidly which results in a sudden spike in frequency. This spike would not occur in real time and results in erroneous tripping in the PSSE model. A WECC white paper found at the hyperlink below explains these limitations in detail.

https://www.wecc.org/Reliability/WECC_White_Paper_Frequency_062618_Clean_Final.pdf

Therefore, relays will be evaluated by inspection of the CONs given in the PSSE .dyr file for the frequency relays. PSSE has an under/over frequency generator bus disconnection relay model, FRQDCAT, and an under/over frequency generator trip relay, FRQTPAT. Both models use the same four CONs as shown in Figure 1 below.

Figure 1: FRQDCAT and FRQTPAT PSSE Model CONs

CONs	Value	Description
J		FL, Lower frequency threshold (Hz)
J+1		FU, Upper frequency threshold (Hz)
J+2		TP, Relay pickup time (sec)
J+3		TB, Breaker time (sec)

Generator Owners requesting to interconnect to the ATC system must not trip within the “no trip zone” for the Eastern Interconnection shown in Attachment 1 of PRC-024-2. The bounds of this “no trip zone” are also shown in Figure 2 below.

Figure 2: PRC-0024-2 Eastern Interconnection Frequency “No Trip Zone” Thresholds

High Frequency Duration		Low Frequency Duration	
Frequency (Hz)	Time (Sec)	Frequency (Hz)	Time (sec)
≥ 61.8	Instantaneous trip	≤ 57.8	Instantaneous trip
≥ 60.5	$10^{(90.935-1.45713*f)}$	≤ 59.5	$10^{(1.7373*f-100.116)}$
< 60.5	Continuous operation	> 59.5	Continuous operation

Continuous Operation Zone

Given these “no trip zone” thresholds, no Generator Owner should have a frequency relay enabled to trip a unit on frequencies between 59.5 Hz and 60.5 Hz. Consequently, FRQDCAT and FRQTPAT models with J above 59.5 or J+1 below 60.5 are considered unacceptable.

Instantaneous Trip Zone

Given these “no trip zone” thresholds, Generator Owners may have frequency relays enabled to trip units at frequencies at or above 61.8 Hz or at or below 57.8 Hz without any relay pickup time. Although Generator Owners are allowed to trip outside of these frequency thresholds, they are not required to trip outside of these frequency thresholds. Consequently, FRQDCAT and FRQTPAT models with J below 57.8 may have any time value in J+2 and be considered acceptable. FRQDCAT and FRQTPAT models with J+1 above 61.8 may have any time value in J+2 and be considered acceptable.

Delayed Trip Zone

Given these “no trip zone” thresholds, Generator Owners may have frequency relays enabled to trip units at frequencies between 60.5 Hz to 61.8 Hz or at frequencies between 59.5 Hz to 57.8 Hz. However, between these frequency thresholds tripping is only allowed after set relay pick times defined by the equations in Figure 2 above. Although Generator Owners are allowed to trip outside of these frequency and relay time thresholds, they are not required to trip outside of these frequency and relay time thresholds. Consequently, FRQDCAT and FRQTPAT models with J between 59.5 and 57.8 must have a time value in J+2 equal to or greater than $10^{(1.7373*f-100.116)}$ to be considered acceptable. FRQDCAT and FRQTPAT models with J+1 between 60.5 and 61.8 must have a time value in J+2 equal to or greater than $10^{90.935-1.45713*f}$ to be considered acceptable.

Evaluating Voltage Protective Relay Settings

Voltage relays will be evaluated in a PSSE dynamics simulation using the power plant .dvr file provided by the Generator Owner. The “no trip zone” voltage thresholds are defined at the Point of Interconnection (POI) and are shown in Figure 3 below.

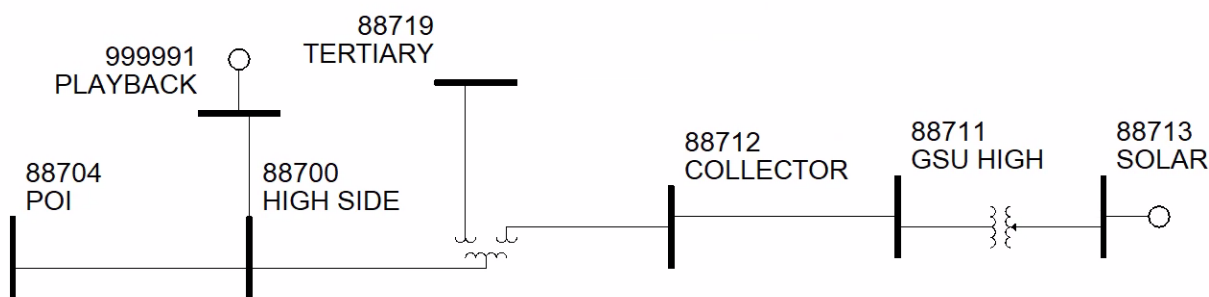
Figure 3: PRC-0024-2 Voltage “No Trip Zone” Thresholds

High Voltage Ride Through Duration		Low Voltage Ride Through Duration	
Voltage (pu)	Time (sec)	Voltage (pu)	Time (sec)
≥ 1.200	Instantaneous trip	< 0.45	0.15
≥ 1.175	0.20	< 0.65	0.30
≥ 1.15	0.50	< 0.75	2.00
≥ 1.10	1.00	< 0.90	3.00

PSSE Model

PRC-0024 defines the “no trip zone” thresholds at the high side of the main power transformer. If the voltage relays are not monitoring the voltage at this location, then it will not be accurate to simply evaluate the parameters of the PSSE model to determine adherence to the PRC-0024-2 standard. Therefore, the best way to assess adherence to the standard is to simulate the high and low voltage thresholds and monitor if any units trip during a simulation where the voltage is set to the PRC-0024-2 voltage thresholds. To perform this simulation a PSSE case modeling the power plant to the POI will need to be created. A PSSE playback generator will be modeled at the main power transformer high side to represent an equivalent for the rest of the system. An example PSSE case with Playback Generator is shown in Figure 4 below.

Figure 4: Example PSSE Model with Playback Generator

Simulation

The playback generator can simulate defined voltage and/or frequency values over time at the bus where the playback machine is located. A playback file must be provided that lists a series of samples of the format: time, voltage, frequency. One simulation should be run with the playback generator inputting the high voltage threshold and another simulation should be run with the playback generator inputting the low voltage threshold. Each simulation will be run for 5 seconds.

If a plant has more than one high voltage relay or more than one low voltage relay each relay should be tested alone. The reason for this is because the different relays will typically have different relay pick up times. If the relay with the shortest pick up time trips it will be impossible to tell if the other relays with longer relay pick up times would have tripped or not.

If any relays trip while running the PRC-0024-2 under or over voltage thresholds the settings will not be considered acceptable.

Evaluating Momentary Cessation

After the simulations have been completed, the active and reactive power of the generator will be plotted. If both active and reactive power are at zero output, at the same time, after the voltage recovers to 0.45 pu during the low voltage simulation then the generator will be deemed to have momentary cessation. Both active and reactive power shall not go to zero, at the same time, during the high voltage simulation.